

Reissner-Mindlin plate

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In this section we discretize and solve the Reissner-Mindlin plate (see [Reissner-Mindlin and Kirchhoff-Love plates](#) for a derivation) with two different methods and discuss the appearance of shear locking. The total energy consisting of bending and shearing reads with the vertical deflection w and the linearized rotation vector β

$$\mathcal{W}_{\text{Reissner-Mindlin}}(w, \beta) = \frac{1}{2} \int_{\Omega} \|\varepsilon(\beta)\|_{\mathbb{D}}^2 + \frac{\kappa G}{t^2} \|\nabla w - \beta\|^2 dx - \int_{\Omega} f \cdot w dx,$$

where t denotes the thickness of the plate, $\varepsilon(\cdot)$ is the symmetric part of the gradient and the elasticity tensor is $\mathbb{D}\varepsilon = \frac{E}{12(1-\nu^2)}((1-\nu)\varepsilon + \nu \operatorname{tr}(\varepsilon)I_{2 \times 2})$ (with E Young's modulus, ν Poisson ratio). Further, $G = \frac{E}{2(1+\nu)}$ and $\kappa = 5/6$ denote the shearing modulus and shear correction factor, respectively.

We are especially interested in positive, but small thicknesses $t > 0$. Therefore, the shear energy acts as a penalty enforcing the equality $\beta = \nabla w$. If there are not enough finite element functions such that the discrete so-called Kirchhoff constraint can be satisfied, we observe bad results known as shear locking.

We consider a benchmark example [[Katili. A new discrete Kirchhoff-Mindlin element based on Mindlin-Reissner plate theory and assumed shear strain fields-Part I: An extended DKT element for thick-plate bending analysis. *International Journal for Numerical Methods in Engineering* 36, 11 \(1993\), 1859-1883](#)] with exact solution. It consists of a circle domain, with clamped boundary and a uniform body force (e.g. gravity) is acting on it. Young modulus E , Poisson ratio ν , and shear correction factor κ .

```
from ngsolve import *
```

```

from netgen.occ import *
from ngsolve.webgui import Draw

E, nu, k = 10.92, 0.3, 5 / 6
G = E / (2 * (1 + nu))
fz = -1

def DMat(mat, E, nu):
    return E / (12 * (1 - nu**2)) * ((1 - nu) * mat + nu * Trace(mat) * Id(2))

def DMatInv(mat, E, nu):
    return (
        12
        * (1 - nu**2)
        / E
        * (1 / (1 - nu) * mat - nu / (1 - nu**2) * Trace(mat) * Id(2))
    )

R = 5

def GenerateMesh(order, maxh=1):
    circ = Circle((0, 0), R).Face()
    circ.edges.name = "circ"
    return Mesh(OCCGeometry(circ, dim=2).GenerateMesh(maxh=maxh * R / 3)).Curve(ord

def ExactSolution(thickness=0.1):
    r = sqrt(x**2 + y**2)
    xi = r / R
    Db = E * thickness**3 / (12 * (1 - nu**2))

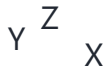
    w_ex = (
        fz
        * thickness**3
        * R**4
        / (64 * Db)
        * (1 - xi**2)
        * ((1 - xi**2) + 8 * (thickness / R) ** 2 / (3 * k * (1 - nu)))
    )
    return w_ex

mesh = GenerateMesh(order=2, maxh=1)
w_ex = ExactSolution(thickness=0.1)
Draw(w_ex, mesh, deformation=True, euler_angles=[-60, 5, 30]);

```

Open Controls

-9.75 -4.88 0.00



NGSolve 6.2.2404-22-g3e237987d

8.4.1. Simple discretization with Lagrange elements

First we consider a simple discretization by using Lagrange elements for the displacements w and rotations β of same polynomial degree k . Then we can directly discretize the Reissner-Mindlin plate equation: Find $(w, \beta) \in \mathbf{Lag}^k \times [\mathbf{Lag}^k]^2$ with $w = \beta = 0$ on $\partial\Omega$ such that for all $(v, \delta) \in \mathbf{Lag}^k \times [\mathbf{Lag}^k]^2$

$$\frac{1}{12} \int_{\Omega} \mathbb{D}\varepsilon(\beta) : \varepsilon(\delta) \, dx + \frac{\kappa G}{t^2} \int_{\Omega} (\nabla w - \beta) \cdot (\nabla v - \delta) \, dx = \int_{\Omega} f v \, dx.$$

```
def SolveReissnerMindlinLagrange(mesh, order=1, thickness=0.1, draw=False):
    fesB = VectorH1(mesh, order=order, dirichlet="circ")
    fesW = H1(mesh, order=order, dirichlet="circ")
    fes = fesB * fesW
    (beta, w), (delta, v) = fes.TnT()

    a = BilinearForm(fes, symmetric=True)
```

```

# bending part
a += InnerProduct(DMat(Sym(Grad(beta))), E, nu), Sym(Grad(delta))) * dx
# shearing part
a += k * G / thickness**2 * InnerProduct(Grad(w) - beta, Grad(v) - delta) * dx
a.Assemble()

f = LinearForm(fz * v * dx).Assemble()

gf_solution = GridFunction(fes)
_, gf_w = gf_solution.components

inv = a.mat.Inverse(fes.FreeDofs(), inverse="sparsecholesky")
gf_solution.vec.data = inv * f.vec

if draw:
    Draw(gf_w, mesh, "w")

return gf_w, fes

```

Solve for different thicknesses for a sequence of refined meshes. Different polynomial order can be tested.

```

results = []
thicknesses = [1, 0.1, 0.01, 0.001]
# try out polynomial order 1 and 2
order = 2

with TaskManager():
    for t in thicknesses:
        results.append([])
        w_ex = ExactSolution(thickness=t)

        for i in range(5):
            mesh = GenerateMesh(order=order, maxh=0.5**i)
            gfw, fes = SolveReissnerMindlinLagrange(
                mesh, order=order, thickness=t, draw=False
            )

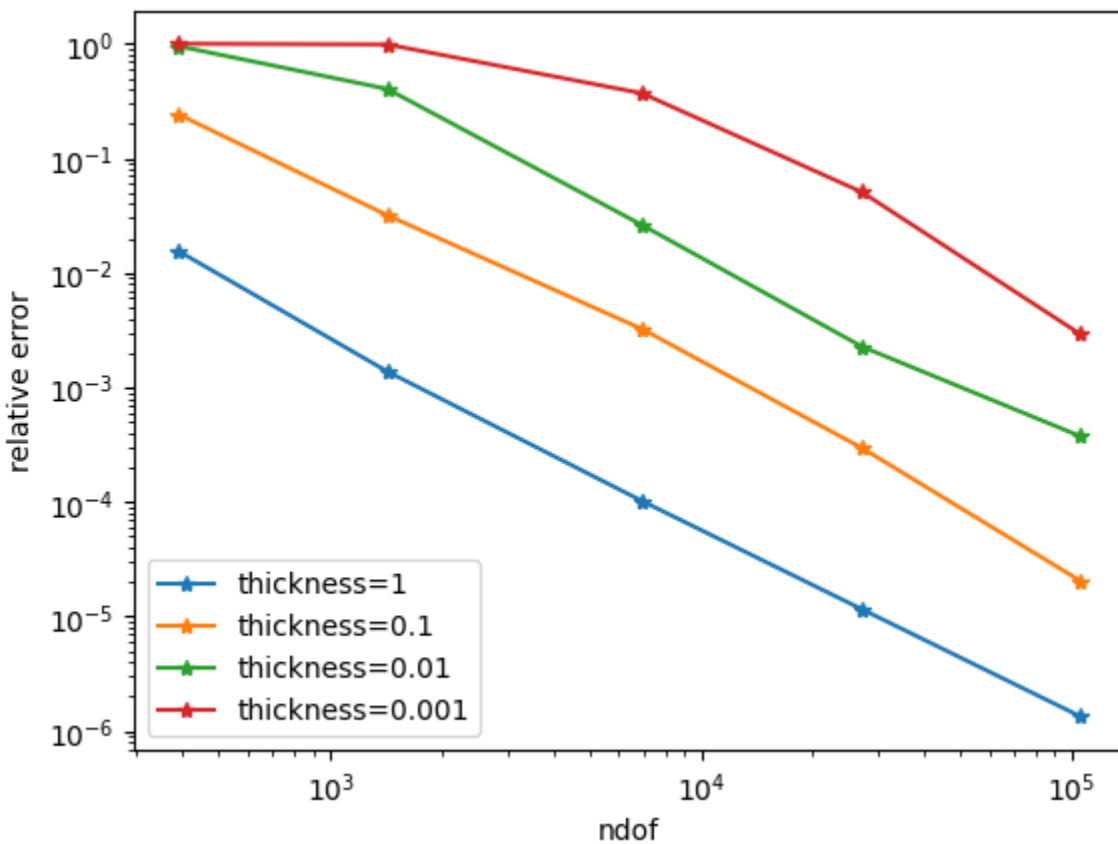
            # relative L2-error of displacement
            err = sqrt(Integrate((gfw - w_ex) * (gfw - w_ex), mesh)) / sqrt(
                Integrate(w_ex * w_ex, mesh)
            )
            results[-1].append((fes.ndof, err))

```

Plot error. We observe immense shear locking for polynomial order $k = 1$ as the pre-asymptotic regime increases rapidly. Increasing the polynomial order to $k = 2$ (or $k = 3$) mitigates the problem, but it is still present.

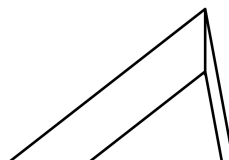
```
import matplotlib.pyplot as plt

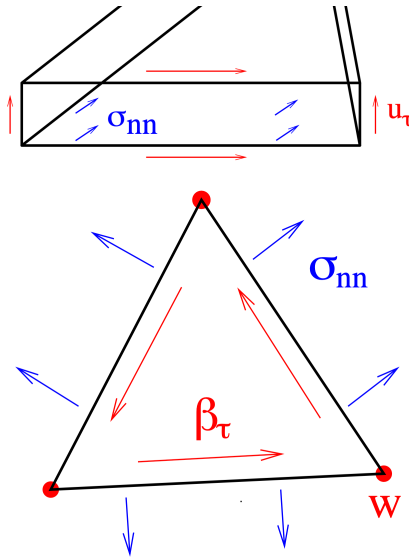
plt.yscale("log")
plt.xscale("log")
plt.xlabel("ndof")
plt.ylabel("relative error")
for i, result in enumerate(results):
    ndof, err = zip(*result)
    plt.plot(ndof, err, "-*", label="thickness=" + str(thicknesses[i]))
plt.legend()
plt.show()
```



8.4.2. TDNNS method

Next, we apply the [TDNNS method](#) to discretize the bending term $\int_{\Omega} \|\varepsilon(\beta)\|_{\mathbb{D}}^2$ as in [\[Pechstein and Schöberl. The TDNNS method for Reissner-Mindlin plates. *Numerische Mathematik* 137, 3 \(2017\), 713-740\]](#).





Instead of using Lagrangian elements, we discretize the rotations β in the Nedelec space. Thus, all discrete gradients ∇w are in the space of rotations, and we don't have the locking problem from before. By introducing the linearized bending moment tensor $\sigma = \mathbb{D}\varepsilon(\beta)$ we obtain the following formulation.

Find $(\sigma, \beta, w) \in H(\text{divdiv}) \times H(\text{curl}) \times H^1(\Omega)$ such that for all $(\tau, \delta, v) \in H(\text{divdiv}) \times H(\text{curl}) \times H^1(\Omega)$

$$\begin{aligned} - \int_{\Omega} \mathbb{D}^{-1} \sigma : \tau \, dx & - \langle \text{div} \tau, \beta \rangle &= 0 \\ - \langle \text{div} \sigma, \delta \rangle & + \frac{\kappa G}{t^2} \int_{\Omega} (\nabla w - \beta)(\nabla v - \delta) \, dx &= \int_{\Omega} f v \, dx, \end{aligned}$$

with the TDNNS duality pairing

$$\begin{aligned} \langle \text{div} \sigma, \beta \rangle &= \sum_{T \in \mathcal{T}} \int_T \text{div} \sigma \cdot \beta \, dx - \int_{\partial T} \sigma_{nt} \beta_t \, ds \\ &= - \sum_{T \in \mathcal{T}} \int_T \sigma : \nabla \beta \, dx + \int_{\partial T} \sigma_{nn} \beta_n \, ds = - \langle \sigma, \nabla \beta \rangle \end{aligned}$$

and the inverted elasticity tensor $\mathbb{D}^{-1} \varepsilon = \frac{12(1-\nu^2)}{Et^3} \left(\frac{1}{1-\nu} \varepsilon - \frac{\nu}{1-\nu^2} \text{tr}(\varepsilon) I_{2 \times 2} \right)$. Here ∂T denotes the element-boundary (edges) of T , n the normal vector and t the tangential vector on ∂T . With e.g. $\sigma_{nt} := t^\top \sigma n$ we denote the normal tangential component of σ .

We use Lagrangian elements of order k for the displacement w , and one order less for the rotations β and the bending moment tensor σ .

```
def SolveReissnerMindlinTDNNS(mesh, order=1, thickness=0.1, draw=False):
```

```

fesB = H Curl(mesh, order=order - 1, dirichlet="circ")
fesS = H DivDiv(mesh, order=order - 1, dirichlet="")
fesW = H1(mesh, order=order, dirichlet="circ")

fes = fesW * fesB * fesS
(w, beta, sigma), (v, delta, tau) = fes.TnT()

n = specialcf.normal(2)

a = BilinearForm(fes, symmetric=True)
# bending part
a += (
    -InnerProduct(DMatInv(sigma, E, nu), tau)
    + InnerProduct(tau, Grad(beta))
    + InnerProduct(sigma, Grad(delta))
) * dx
a += -(sigma[n, n] * delta * n + tau[n, n] * beta * n) * dx(element_boundary=True)
# shearing part
a += k * G / thickness**2 * InnerProduct(Grad(w) - beta, Grad(v) - delta) * dx
a.Assemble()

f = LinearForm(fz * v * dx).Assemble()

gf_solution = GridFunction(fes)
gf_w, _, _ = gf_solution.components

inv = a.mat.Inverse(fes.FreeDofs(), inverse="")
gf_solution.vec.data = inv * f.vec
if draw:
    Draw(gf_w, mesh, "w")

return gf_w, fes

```

```

results = []

# try out polynomial order 1 and 2
order = 1

thicknesses = [1, 0.1, 0.01, 0.001]
with TaskManager():
    for t in thicknesses:
        results.append([])
        w_ex = ExactSolution(thickness=t)

        for i in range(5):
            mesh = GenerateMesh(order=order, maxh=0.5**i)
            gf_w, fes = SolveReissnerMindlinTDNNS(
                mesh, order=order, thickness=t, draw=False
            )

            # relative L2-error of displacement
            err = sqrt(Integrate((gf_w - w_ex) * (gf_w - w_ex), mesh)) / sqrt(
                Integrate(w_ex * w_ex, mesh)
            )

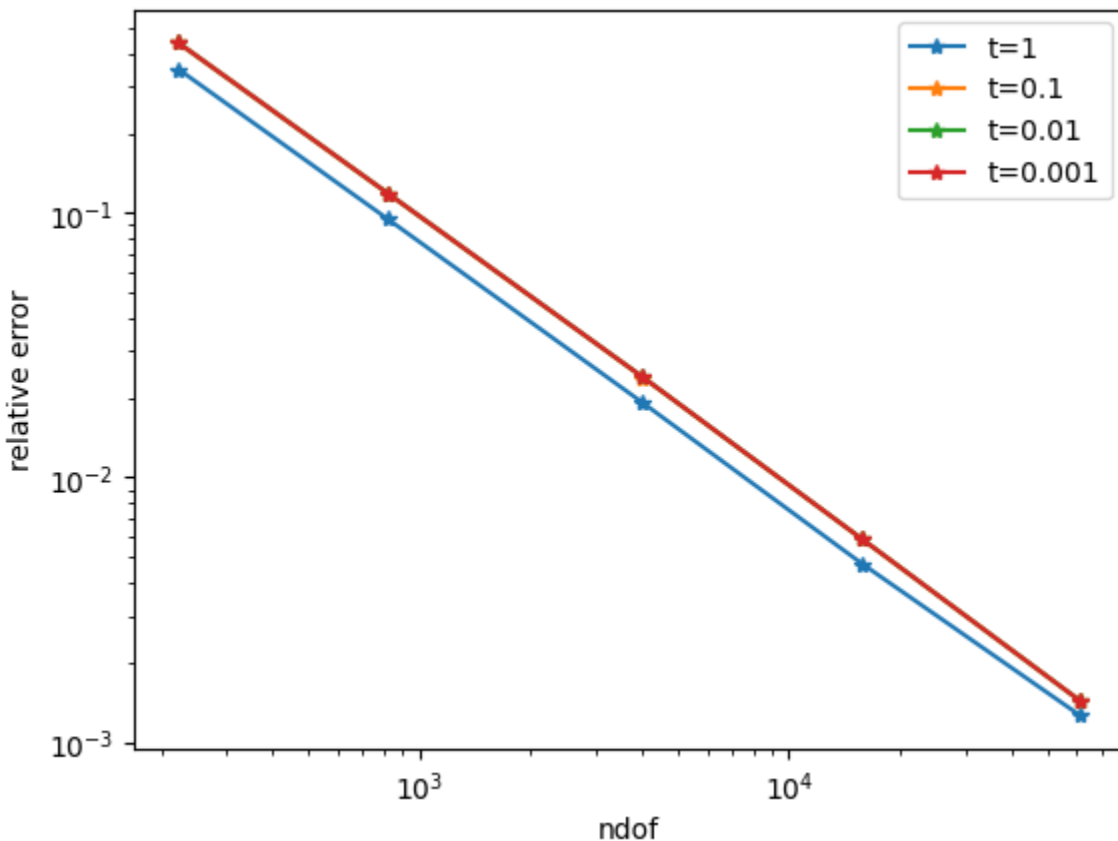
```

```
)
results[-1].append((fes.ndof, err))
```

With the TDNNS method we obtain a locking free discretization already for polynomial degree $k = 1$ for the displacements.

```
import matplotlib.pyplot as plt

plt.yscale("log")
plt.xscale("log")
plt.xlabel("ndof")
plt.ylabel("relative error")
for i, result in enumerate(results):
    ndof, err = zip(*result)
    plt.plot(ndof, err, "-*", label="t=" + str(thicknesses[i]))
plt.legend()
plt.show()
```



In the limit $t \rightarrow 0$ the shear energy term can be understood as a penalty formulation enforcing the Kirchhoff-Love assumption

$$\beta = \nabla w.$$

Thus, in the limit case, the total energy simplifies by eliminating the rotation β to

$$\mathcal{W}_{\text{Kirchhoff-Love}}(w) = \frac{1}{2} \int_{\Omega} \|\varepsilon(\nabla w)\|_{\mathbb{D}}^2 - \int_{\Omega} f w,$$

which is the Kirchhoff-Love plate model.

Drawback of the current implementation of the TDNNS formulation for Reissner-Mindlin plates: We have a mixed saddle point problem leading to an indefinite system matrix. This can be easily overcome with hybridization techniques [NGSolve docu - Static condensation](#) by breaking the normal-normal continuity of σ and reinforcing it by a Lagrange multiplier α . After static condensation a symmetric positive definite system is regained.

Exercise:

- Implement hybridization for TDNNS for Reissner-Mindlin plates

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